Práctica 3: Sistema musculoesqueletico

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Información general

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Asignatura: **Modelado de Sistemas Fisiológicos**

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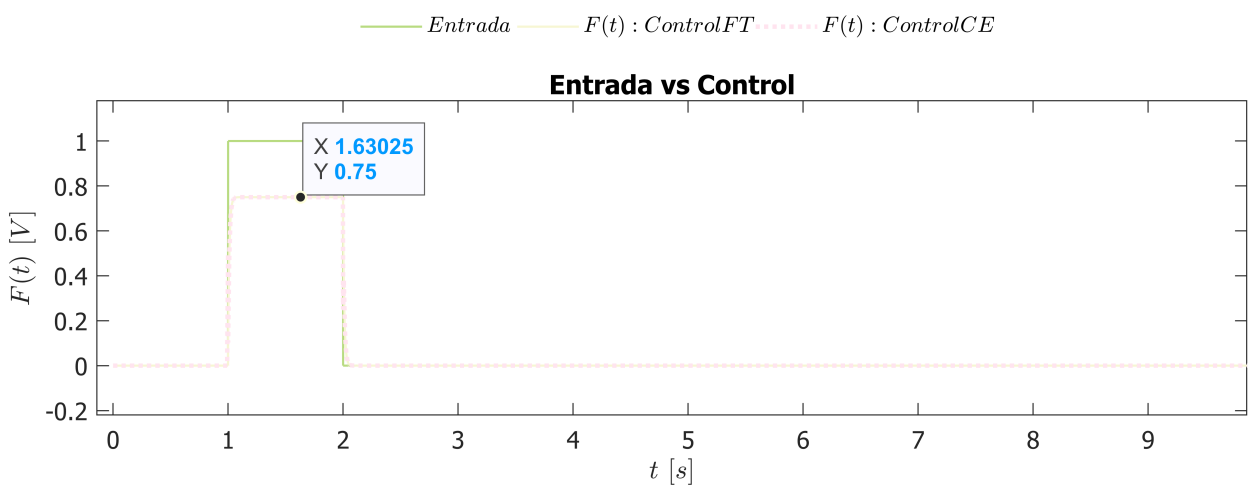
Datos de la simulación

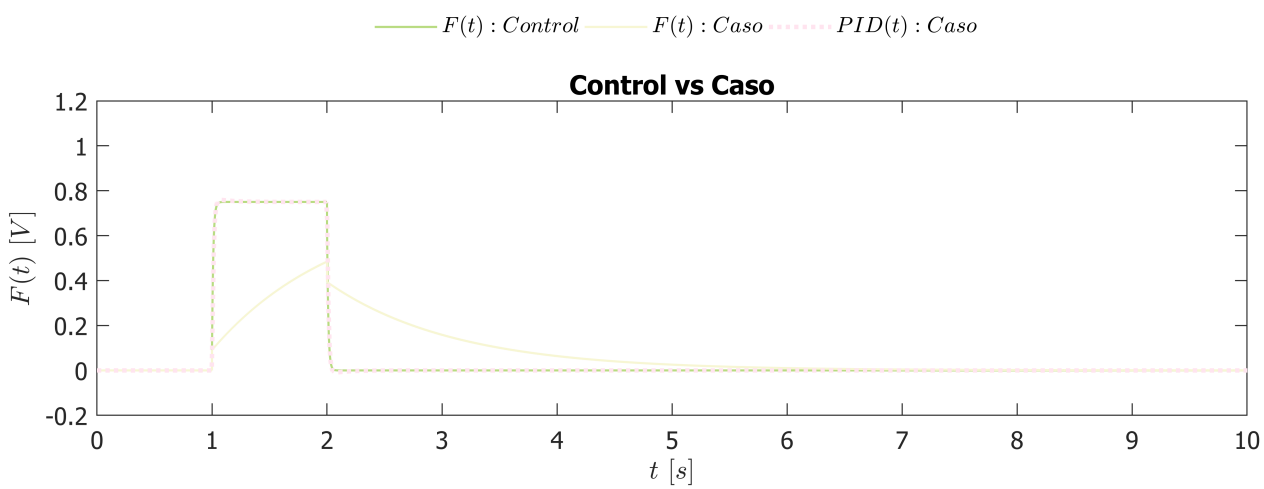
```
clc; clear; close all; warning('off','all')
tend = '10';
file = 'Sistema';
open_system(file);
parameter.StopTime = '15';
parameter.Solver = 'ode15s';
parameter.MaxStep = '1E-3';
x = sim(file,parameter);
```

Respuesta del sistema

```
plotsignals(x.t,x.F,x.Fs1,x.Fs2,x.Fs3,x.PID)
```

```
subplot(2,1,1)
xlim([-0.14 9.86])
ylim([-0.22 1.18])
ax = gca;
chart = ax.Children(1);
datatip(chart,1.63,0.75);
```





Funcion de las señales

```
function plotsignals(t,F,Fs1,Fs2,Fs3,PID)
set(figure(),'Color','w')
set(gcf,'units','centimeters','position',[1,1,25,20])
set(gca,'FontName','Times New Roman','FontSize',11)
hold on; grid off; box on;
powerrangers = [184, 219, 128;
                247, 246, 211;
```

```
                 255, 228, 239]/255;
colororder(powerrangers)
subplot(2,1,1);box on; hold on;

P = plot(t,F,'-',t,Fs1,'-',t,Fs2,':','LineWidth',1); set (P(3),'LineWidth',2)
L = legend('$Entrada$','$F(t):Control FT$','$F(t):Control CE$');
set(L,'Interpreter','Latex','FontSize',10,'location',...
'NorthOutside','box','off','Orientation','Horizontal')
xlabel('$t$ $[s]$','Interpreter','Latex','FontSize',11)
ylabel('$F(t)$ $[V]$','Interpreter','Latex','FontSize',11)
xlim([0,10]); xticks(0:1:10);
ylim([-0.2,1.2]); yticks(-0.2:0.2:1.2)
T = title("Entrada vs Control"); set(T,'FontSize',10);
set(gca,'FontName','Time News Roman','FontSize',11);
subplot(2,1,2);box on; hold on;
P = plot(t,Fs1,'-',t,Fs3,'-',t,PID,':','LineWidth',1); set (P(3),'LineWidth',2)
L = legend('$F(t):Control$','$F(t):Caso$','$PID(t):Caso$');
set(L,'Interpreter','Latex','FontSize',10,'location',...
'NorthOutside','box','off','Orientation','Horizontal')
xlabel('$t$ $[s]$','Interpreter','Latex','FontSize',11)
ylabel('$F(t)$ $[V]$','Interpreter','Latex','FontSize',11)
xlim([0,10]); xticks(0:1:10);
ylim([-0.2,1.2]); yticks(-0.2:0.2:1.2)
T = title("Control vs Caso"); set(T,'FontSize',10);
set(gca,'FontName','Time News Roman','FontSize',11)

drawnow
exportgraphics(gcf,'musculoesqueletico.pdf','ContentType','vector')
end
```